New Keynesian Models: Not Yet Useful For Policy Analysis

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Summary

- Critique of the "structural" shocks, as in Smets and Wouters (2007) (SW), mainly the wage markup shocks
- SW's shock arises due to fluctuations in the elasticity of substitution across different types of labor
- In units of a markup, the shock has a mean of 50 percent and a standrad deviation of over 2500 percent (absurd)
- CKM show that this wage markup is equivalent to a labor wedge, which can have many interpretations
- CKM simply show that the SW model is not useful for policy as its shocks are neither
  1. policy invariant
  2. interpretable

Other dubious shocks

Three of the seven exogenous random variables are arguably strctural

- shocks to total factor productivity
- investment-specific technology
- monetary policy

The remaining four shocks are dubiously structural:

- wage markups
- price markups
- exogenous spending
- risk premia

TABLE 1—FORECAST ERROR VARIANCE DECOMPOSITION IN THE SMETS-WOUTERS (2007) MODEL

		Each shock's percent contribution to variance of		
Horizon	Type of Shock	Output	Hours	Inflation
4 Quarters	1–Monetary shock	9.1	11.1	4.6
	2–Productivity shock	26.1	8.1	5.1
	3–Investment shock	25.1	26.8	3.3
	4–Risk premium shock	9.9	12.8	0.7
	5–Exogenous spending shock	15.2	20.9	0.5
	6–Price markup shock	26.1	8.1	5.1
	7–Wage markup shock	6.5	12.9	42.9
	All shocks	100.0	100.0	100.0
	Shocks 4–7	39.6	53.9	86.9
10 Quarters	1–Monetary shock	5.9	8.0	5.3
	2–Productivity shock	31.6	4.1	4.5
	3–Investment shock	18.5	18.7	3.6
	4–Risk premium shock	4.2	6.3	0.7
	5–Exogenous spending shock	8.2	13.8	0.7
	6–Price markup shock	11.1	11.4	34.0
	7–Wage markup shock	20.5	37.7	51.1
	All shocks	100.0	100.0	100.0
	Shocks 4–7	44.0	69.2	86.5
1,000 Quarters	1–Monetary shock	2.3	3.4	4.6
	2–Productivity shock	29.5	2.0	4.0
	3–Investment shock	7.9	8.6	3.4
	4–Risk premium shock	1.6	2.6	0.6
	5–Exogenous spending shock	4.2	10.5	1.0
	6–Price markup shock	6.4	6.2	28.6
	7–Wage markup shock	48.2	66.7	57.8
	All shocks	100.0	100.0	100.0
	Shocks 4–7	60.3	86.0	88.0

Note: Output and hours are logged and detrended but not differenced.

Prototype Growth Model Setup

They begin with the model with four reduced-form shocks, referred to as wedges:

- A_t = efficiency wedge, $1 - \tau_t^l$ = labor wedge, $1/(1 + \tau_t^x)$ = investment wedge, g_t = government wedge

Consumers maximize:

$$\max_{\{c_t, l_t, x_t, k_{t+1}\}} E_0 \sum_{t=0}^{\infty} \beta^t U(c_t, 1 - l_t)$$

subject to the budget constraint:

$$c_t + (1 + \tau_t^x) x_t = (1 - \tau_t^l) w_t l_t + r_t k_t + T_t$$

and the capital accumulation law:

$$k_{t+1} = (1 - \delta) k_t + x_t$$

Resource constraint:

$$c_t + x_t + g_t = y_t$$

Equilibrium

Technology equation:

$$y_t = A_t F(k_t, l_t)$$

Labor wedge condition:

$$\frac{U_{l_t}}{U_{c_t}} = (1 - \tau_t^l) A_t F_{l_t} \equiv \frac{U_{l_t}}{U_{c_t}} = (1 - \tau_t^l) w_t$$

Euler equation:

$$U_{c_t}(1 + \tau_t^x) = E_t\Big[\beta U_{c,t+1}\big(A_{t+1} F_{k,t+1} + (1-\delta)(1 + \tau_{t+1}^x)\big)\Big]$$

How important is the labor wedge?

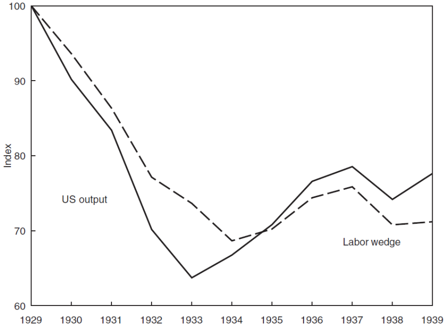

FIGURE 1A. US OUTPUT AND THE MEASURED LABOR WEDGE
(Annual 1929–39, Series Normalized to Equal 100 in 1929)

Source: Chari, Kehoe, and McGrattan (2007)

FIGURE 1B. US LABOR AND PREDICTION OF MODEL WITH JUST THE LABOR WEDGE
(Annual 1929–39, Series Normalized to Equal 100 in 1929)

Source: Chari, Kehoe, and McGrattan (2007)

Fluctuating Government Policy Toward Unions

Technology producing final goods:

$$y(s^t) = F(k(s^{t-1}, l(s^t)))\text{where } s^t\text{is a history of exogenous shocks}$$

An aggregate of the differentiated types of labor:

$$l(s^t) = \left(\int_0^1 l(i, s^t)^{\frac{1}{1+\lambda}}\, di\right)^{1+\lambda}$$

where λ governs an elasticity of substitution. Then the only distorted FOC:

$$w(i, s^t) = (1+\lambda)\frac{u_l(i, s^t)}{u_c(i, s^t)},\ w(s^t) = [1+\bar{\lambda}(s^t)]\frac{u_l(s^t)}{u_c(s^t)},\ \text{where}\bar{\lambda}(s^t) \leq \lambda$$

In the prototype economy:

$$[1-\tau(s^t)]w(s^t) = \frac{u_l(s^t)}{u_c(s^t)},\ \text{therefore } [1-\tau(s^t)] = \frac{1}{1+\bar{\lambda}(s^t)}$$

Fluctuating Utility of Leisure

Now consumers in the economy have the following utility:

$$u(c(s^t), 1 - l(s^t)) = u(c(s^t)) + \psi(s^t)v(1 - l(s^t))$$

where ψ_t is an exogenous stochastic shock to the utility of leisure v, and the FOC is given by:

$$\frac{v'(1 - l(s^t))}{u'(c(s^t))} = \frac{w(s^t)}{\psi(s^t)}$$

In the prototype economy we have the following utility function:

$$u(c(s^t), l(s^t)) = u(c(s^t)) + v(1 - l(s^t))$$

with the consumer's FOC:

$$\frac{v'(1 - l(s^t))}{u'(c(s^t))} = [1 - \tau(s^t)]w(s^t), \text{ therefore:}$$

$$1 - \tau(s^t) = \frac{1}{\psi(s^t)}$$

SW Model and critique - a wage markup is equivalent to a labor wedge

Labor aggregator G relates aggregate labor l_t to a continuum of differentiated types of labor services $l_t(i)$, according to:

$$1 = \int_0^1 G\left(\frac{l_t(i)}{l_t}; \lambda_t\right) di$$

where λ_t is referred to as a wage markup shock.

$$l_t = \left(\int_0^1 l_t(i)^{\frac{1}{1+\lambda_t}}\, di\right)^{1+\lambda_t} \qquad \text{(CES special case)}$$

To see how the wage markup shock is equivalent to the labor wedge,

$$w_t = (1+\lambda_t)\frac{u_{l_t}}{u_{c_t}}, \text{ or } \frac{u_{l_t}}{u_{c_t}} = \frac{1}{1+\lambda_t} w_t$$

Wage markup shock = labor wedge

$$\frac{U_{l_t}}{U_{c_t}} = (1-\tau_t)w_t$$

$$\text{Union interpretation:} \qquad 1-\tau_t = \frac{1}{1+\bar{\lambda}_t}$$

$$\text{Leisure interpretation:} \qquad 1-\tau_t = \frac{1}{\psi_t}$$

$$\text{Smets–Wouters wage markup:} \qquad 1-\tau_t = \frac{1}{1+\lambda_t}$$

- They reestimate the SW model for the CES case after imposing 50 percent mean markup, standard deviation 2587 percent.
- In SW these fluctuations correspond to fluctuations in the elasticity of substitution between different types of labor
- Interpretation - shock may stand in for some deeper yet unidentified

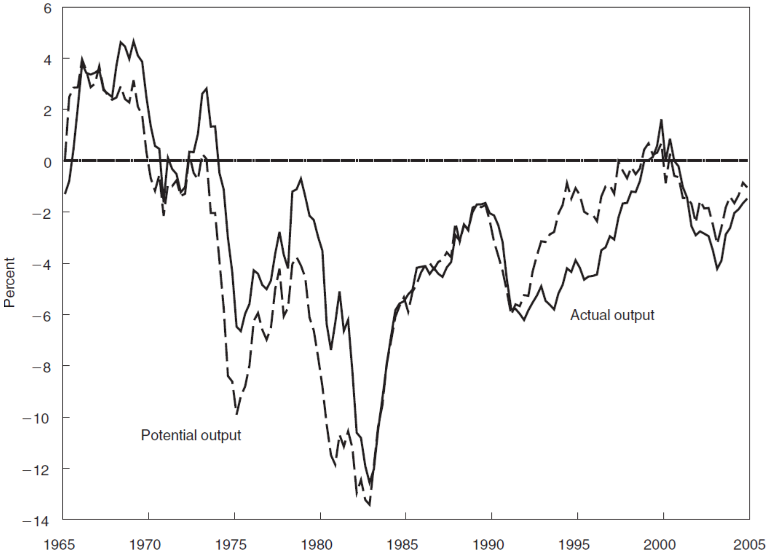

FIGURE 3. US OUTPUT AND POTENTIAL OUTPUT IN VERSION OF SMETS-WOUTERS (2007) MODEL WITH AR(1) TASTE SHOCKS AND INDEPENDENTLY AND IDENTICALLY DISTRIBUTED WAGE MARKUP SHOCKS
(Quarterly Percentage Changes, 1965–2005, Series Logged and Detrended)